

KARTHIK BHARADWAJ

Lead Data Scientist

Accomplished Machine Learning Engineer with a proven track record of designing, developing, and deploying high-performance AI/ML models for computer vision, NLP, and multimodal systems. Known for the ability to break down complex problems and bring cross-functional teams together to deliver scalable, production-grade solutions that drive measurable business value. Adept at translating business requirements into innovative AI products under tight deadlines, leveraging state-of-the-art technology and methodologies. Recognized for exceptional problem-solving skills and a commitment to continuous learning and innovation in the field of artificial intelligence.



WORK EXPERIENCE

- Nielsen Media Private Limited** MAY 2025 - PRESENT
Lead Data Science Engineer
Led multiple AI/ML projects, including developing an agent-based automated research pipeline to improve editor productivity by autonomously gathering, validating, and populating relevant program information. Additionally built a cross-encoder improved the previously XGBoost-based matching system to map sparsely or vaguely worded program data to accurate TMS IDs, significantly enhancing data quality and retrieval accuracy.
- Nference Labs Private Limited** MAR 2023 - MAY 2025
Lead Data Scientist
Responsible for spearheading the development of cutting-edge AI solutions in the healthcare domain. I led teams in building innovative systems for cardiac mapping and arrhythmia detection, utilizing impedance-based technology and machine learning techniques to optimize diagnostic accuracy and efficiency.
- Statestreet Corporation** AUG 2021 - MAR 2023
Emerging Lead - AI/ML Engineer
Led the AI India initiatives for a prominent fintech client, driving business growth through advanced machine learning solutions. My work focused on optimizing fund benchmark predictions, enhancing predictive model accuracy, and improving deployment efficiency. I collaborated closely with stakeholders to translate business requirements into actionable technical solutions, ensuring that AI-driven innovation delivered measurable business value.
- Dimaag-AI Private Limited** DEC 2020 - AUG 2021
Senior AI/ML Engineer
At Dimaag AI, I played a pivotal role in architecting and implementing AI-driven solutions tailored to the needs of global clients. I worked closely with the CTO to define the technical roadmap and ensure that the solutions met both business and engineering requirements. My role spanned from designing scalable AI architectures to hands-on implementation, optimizing models for real-world deployment. Additionally, I handled direct client communications, aligning technical solutions with business objectives and demonstrating model capabilities to stakeholders.
- Wipro Limited** DEC 2019 - DEC 2020
Senior AI/ML Engineer
Worked on the development of Wipro's proprietary Explainable AI (XAI) solution, E.T.H.I.C.A, which focuses on ensuring AI transparency and interpretability. I worked on designing and implementing the database architecture supporting the explainability features and developing key metrics for evaluating the solution. Additionally, I contributed to CIP/DEFT, an advanced invoice processing pipeline, leveraging machine learning techniques to extract critical financial entities with high accuracy.
- Ambertag Analytics Private limited** JUN 2018 - DEC 2019
Data Analyst / Data Engineer
As a Data Analyst/Data Engineer at Ambertag, I worked on integrating and analyzing large datasets for e-commerce and home interior clients. My role involved designing and implementing systems to streamline data flow, enabling efficient processing and insightful business analysis. I leveraged machine learning techniques to optimize product recommendations and analyzed client data to derive actionable insights, ensuring data accuracy and supporting business decisions through exploratory data analysis (EDA) and scalable data management solutions.
- Lowe's India Services Private Limited** AUG 2017 - MAY 2018
Associate Software Engineer
At Lowe's, I worked on resolving ad hoc issues and bug fixes as an Associate Software Engineer. I was responsible for developing software solutions and ensuring smooth data flow across systems. I gained hands-on experience with webMethods, a critical integration platform, as well as contributed to the development of web-based solutions for various internal tools.

EDUCATION

2013 - 2017

RNS Institute of Technology

- Bachelor of Engineering

SKILLS & FRAMEWORKS

- Python
- SQL
- Linux
- Docker
- Exploratory Data Analysis(EDA)
 - Numpy
 - Pandas
- Data Engineering
- Data Visualization
 - Matplotlib
 - Plotly
- Machine Learning
 - Scikit-learn
 - Scipy
- Deep Learning
 - Pytorch
 - Keras
 - Tensorflow
- Natural Language Processing(NLP)
 - NLTK
 - Transformers
- Image Processing/Computer Vision(CV)
 - OpenCV
 - PIL
 - CNN, GANs
 - Auto-Encoders
- Large Language Models(LLMs)
- Generative AI
- Cloud Platforms
 - AWS
 - GCP
 - Azure

EXTRA-CURRICULAR

Has been a student of Carnatic music for the past 17 years
Graded Artist of All India Radio in the category of **Carnatic Flute**.

PROJECTS

● **IMPEDANCE** - *Nference Labs Private Limited*

Developed a next-generation, impedance-based cardiac mapping system (EAM), revolutionizing catheter guidance by eliminating dependence on proprietary magnetic systems—significantly reducing both cost and system complexity.

- **System Architecture & Real-Time Data Processing:** Designed and implemented a high-performance data acquisition and processing pipeline with near-zero latency, ensuring seamless real-time communication between hardware and software while optimizing storage efficiency without impacting runtime.
- **Machine Learning & Algorithm Optimization:** Leveraged advanced machine learning techniques to enhance the accuracy and efficiency of electro-anatomical maps, refining signal processing methodologies and boosting diagnostic precision.
- **Performance & Accuracy:** Achieved an industry-leading root mean square error (RMSE) of ~2mm compared to traditional magnetic mapping techniques, validating the system's reliability for precise cardiac diagnostics.
- **Innovation & Cost Reduction:** Pioneered an alternative to expensive and proprietary magnetic systems, making cardiac mapping more accessible while maintaining clinical-grade accuracy, positioning the system as a cost-effective solution in electrophysiology.

This role required deep expertise in real-time signal processing, medical imaging, and machine learning, coupled with strong problem-solving and system design skills to push the boundaries of cardiac electrophysiology.

● **SIG-ID** - *Nference Labs Private Limited*

Designed and implemented a deep learning-based system for identifying Pulmonary Vein Potential (PVP) signatures in ECG signals, addressing the critical challenge of distinguishing between visually similar patterns, such as pacing spikes and other recurrent but random signals.

- **Advanced Signal Processing & Pattern Recognition:** Developed a robust framework to accurately detect PVP signatures, overcoming noise and signal overlap that traditionally hinder precise arrhythmia mapping.
- **Deep Learning Implementation:** Leveraged UNET-1D, a specialized deep learning architecture for one-dimensional signal segmentation, to precisely isolate and classify PVPs from complex ECG waveforms.
- **High-Accuracy Performance:** Achieved an exceptional 98% classification accuracy, significantly enhancing the reliability of arrhythmia diagnostics and improving outcomes in electrophysiological procedures.
- **Impact on Electrophysiology:** Improved the precision of PVP detection, enabling more effective pulmonary vein isolation (PVI) procedures—critical in the treatment of atrial fibrillation and other arrhythmias.

This project combined deep learning, biomedical signal processing, and domain-specific expertise to advance the accuracy and effectiveness of ECG-based arrhythmia diagnostics.

● **NEUVIZ** - *Nference Labs Private Limited*

Designed and enhanced a post-procedural analysis tool for electro-anatomical mapping, providing physicians with advanced visualization and data-driven insights to improve follow-up treatment strategies and patient outcomes.

- **Seamless Data Integration & Visualization:** Developed a robust system to load, process, and visualize electro-anatomical maps from both proprietary and third-party mapping systems, enabling direct side-by-side comparisons for enhanced procedural assessment.
- **Advanced Metrics & Clinical Insights:** Augmented existing data with additional computed metrics, offering clinicians a deeper understanding of cardiac geometry, lesion effectiveness, and procedural outcomes beyond the raw mapping data.
- **Enhanced Decision Support:** Enabled physicians to retrospectively analyze historical procedure data, facilitating better-informed decisions for follow-up treatments and refinement of ablation strategies.
- **Impact on Electrophysiology Workflow:** Improved accessibility and interpretability of electro-anatomical data, reducing reliance on fragmented or incomplete records and enhancing continuity of care in electrophysiology procedures.

This tool significantly advanced post-procedural evaluation by integrating multi-source data and providing novel insights, ultimately helping physicians optimize treatment plans for better patient outcomes.

● **DYNAMIC BENCHMARKING** - *Statestreet Corporation*

Developed a dynamic benchmarking model to enhance the predictive accuracy of fund and securities valuation by analyzing historical relationships with market indices.

- **Feature Selection & Model Optimization:** Addressed the challenge of selecting the most relevant indices from a dataset of over 5,000 options by leveraging Lasso regression for feature selection, effectively isolating the top 50 indices with the highest impact on fund movements.
- **Predictive Modeling for Fund Valuation:** Built and deployed a high-performance predictive model capable of estimating fund and security values with superior accuracy, significantly improving risk assessment and financial decision-making.
- **Production Deployment & Business Impact:** Successfully deployed the model in a production environment, where it has outperformed the business benchmark in over 70% of cases, enhancing fund performance predictions and valuation reliability.
- **Strategic Financial Insights:** Provided asset managers with a data-driven approach to assess fund performance, leading to more informed investment strategies and risk mitigation.

This solution optimized fund valuation processes, leveraging machine learning to drive better financial insights and improve decision-making in asset management.

● **WAREHOUSE PPE DETECTION** - *Dimaag AI Private Limited*

Developed an AI-driven PPE Detection system for a Japanese manufacturing client, enhancing workplace safety by ensuring compliance with protective equipment requirements through real-time CCTV monitoring.

- **Computer Vision for Workplace Safety:** Designed and implemented a robust object detection model to identify essential personal protective equipment (PPE)—including vests, goggles, gloves, and hard hats—using warehouse surveillance footage.
- **Optimized Deep Learning Model:** Leveraged Tiny YOLOv4, a lightweight yet high-performance object detection model, to achieve efficient real-time inference with an average per-class accuracy of 95% on client-specific data.
- **Scalable & Reusable AI Components:** Developed modular, reusable components to streamline the training and deployment of custom object detection models, enabling rapid adaptation to new PPE requirements with minimal manual intervention.
- **Impact on Workplace Safety & Compliance:** Provided a scalable, AI-powered solution that improved monitoring efficiency, reduced compliance violations, and ensured a safer working environment for warehouse employees.

PROJECTS

BALL GRID ARRAY - Dimaag AI Private Limited

Developed an advanced BGA Transfer system for a leading chip fabrication client, ensuring precise and reliable solder ball transfers while maintaining critical semiconductor manufacturing parameters.

- **High-Precision Image Processing:** Designed and implemented a robust solution to analyze high-resolution 3D ball grid array (BGA) images, leveraging traditional image processing techniques to accurately detect, segment, and track individual solder balls.
- **Advanced Image Blending for Seamless Integration:** Applied Poisson Blending, a sophisticated image fusion technique, to seamlessly integrate transferred solder balls into the target grid, preserving both structural integrity and electrical properties.
- **End-to-End Solution Architecture & Implementation:** Led the entire solution design, collaborating closely with the client to define requirements, validate outputs, and ensure compliance with stringent semiconductor industry standards. Independently implemented and optimized the solution for accuracy, efficiency, and reliability.
- **Impact on Semiconductor Manufacturing:** Delivered a high-precision system that ensured minimal deviation in key parameters—including noise profile, brightness, volume, and size—significantly enhancing chip fabrication quality and consistency.

This project showcased expertise in image processing, computer vision, and semiconductor manufacturing, delivering a cutting-edge solution that met the industry's rigorous precision standards.

COGNITIVE IMAGE PROCESSING/DEEP EXTRACTION FRAMEWORK FOR TEXT(CIP/DEFT) and XAI - Wipro Limited

Designed and developed an intelligent invoice processing pipeline and contributed to Wipro's proprietary Explainable AI (XAI) solution, enhancing automation and interpretability in AI-driven decision-making.

- **Automated Invoice Processing:** Built an advanced pipeline to extract key financial entities—including Total Amount, GST, Vendor ID, Date, and Address—from both image-based and text-based PDFs.
- **OCR & NER Implementation:** Integrated preprocessing techniques, Optical Character Recognition (OCR), and a Named Entity Recognition (NER) model, achieving 85–90% OCR accuracy and 95% NER accuracy for precise data extraction.
- **Contribution to Explainable AI (XAI):** Played a key role in Wipro's E.T.H.I.C.A (Explainable AI) initiative by designing the supporting database architecture and defining key metrics to evaluate Explainability, Transparency, Human-Centric Design, Interpretability, Common Sense, and Auditability.
- **Business Impact:** Enhanced financial document automation, reducing manual effort in invoice processing while ensuring compliance and auditability through explainable AI frameworks.

This work demonstrated expertise in NLP, OCR, and explainable AI, driving automation and transparency in enterprise AI solutions.

MATERIAL - RECOMMENDATION ENGINE - AmberTag Analytics Private Limited

Developed a scalable recommendation system for Material, a fashion e-commerce client, by integrating deep learning and data engineering techniques to enhance product discovery and personalization.

Project Scope & Challenges:

The project aimed to streamline product recommendations by leveraging multiple data sources, ensuring a structured and consistent dataset for model training. The primary challenges included:

1. **Populating Product Feeds:** Many items lacked recommendations due to incomplete metadata. To address this, an image classification pipeline was developed to automatically categorize and label products.
2. **Fetching Similar Products:** Extracted visual attributes from images using deep learning models to recommend visually similar products, enhancing customer engagement and personalization.

Deep Learning Model Development & Implementation

- Designed and trained four deep learning models for attribute extraction and image classification, achieving 85–90% test accuracy across different fashion attributes.
- Optimized model architectures for efficiency, ensuring real-time performance in a production environment.
- Integrated structured and unstructured data sources (images, metadata, and customer interactions) to refine recommendations.

Deployment & Business Impact

- Built a minimalistic CLI-based recommendation system, enabling quick retrieval of similar products and category-based suggestions.
- Successfully delivered one of AmberTAG's first major deep learning client projects, setting a foundation for future AI-driven initiatives in fashion e-commerce.
- Improved recommendation accuracy, leading to better product visibility and enhanced customer experience, ultimately driving higher engagement and potential sales growth.

INVENTIONS(PATENTS AND IPS)

- **1518-091USU1:** Apparatus and method for generating clinical decision support
- **1518-092USU1:** Apparatus and method for correcting machine learning model predictions
- **1518-093USP1:** Method and system for labeling of intracardiac signals
- **1518-093USU1:** Apparatus and a method for a plurality of time series data
- **1518-094USU1:** Apparatus and method for determining a label dynamically for a potential signal
- **1518-095USP1:** Semi-supervised machine learning model
- **1518-096USP1:** System and method for the intelligent detection, labeling, and visualization of cardiac signals enabled by medical case data analysis
- **1518-097USP1:** Data management and machine learning models for labelling of cardiac signal
- **1518-185USU1:** Apparatus and method for determining a normalized voltage across non-identical electrodes
- **1518-186USU1:** Systems and methods for artificial intelligence assisted medical device localization
- **1518-189USU1:** Methods and systems for determining position signals of electrodes using a retrained machine-learning model
- **US-12198572-B1:** Apparatus and Method for Generating an Impedance Model of a Biological Chamber